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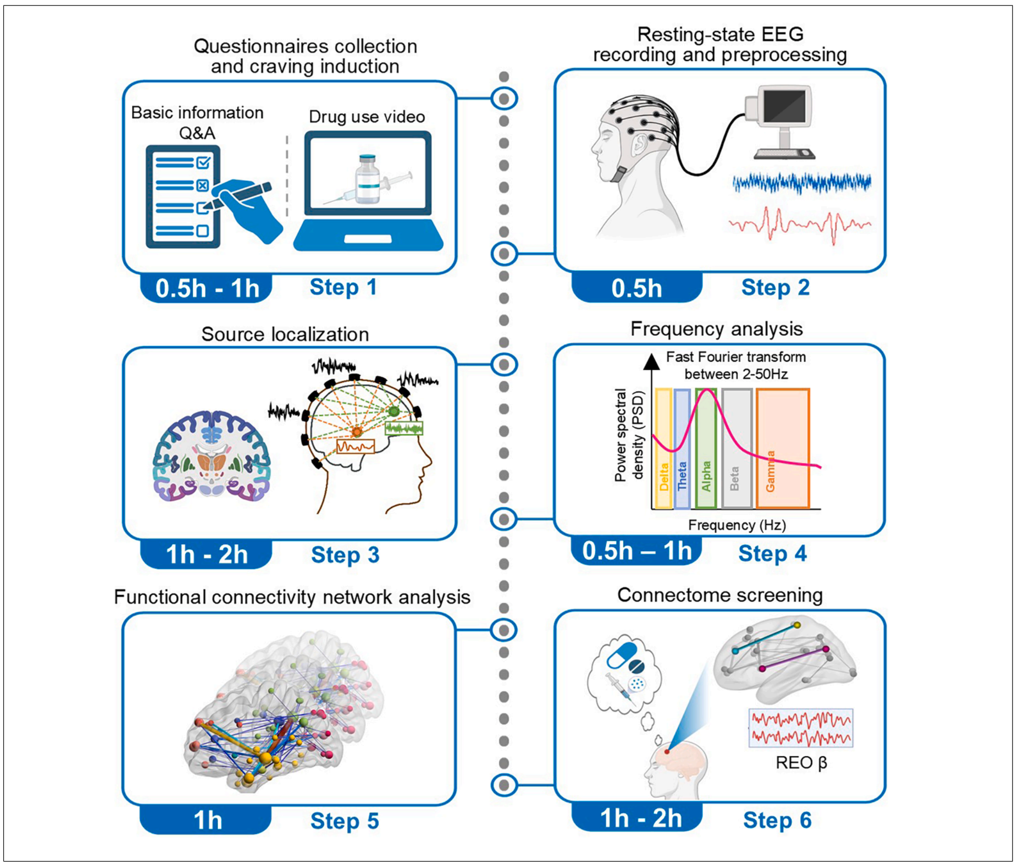
# Protocol to investigate neurophysiological link between resting-state brain activity and clinical symptoms in methamphetamine use disorder




The absence of reliable biomarkers impedes the objective diagnosis and monitoring of many psychiatric symptoms. Here, using cue-induced craving in methamphetamine use disorder as a case example, we present a protocol for linking neurophysiological characteristics to clinical symptoms through high-density electroencephalography (HD-EEG). We describe steps for HD-EEG data acquisition, signal preprocessing, source localization, and functional connectivity network construction. We then detail procedures for association analysis with clinical symptom scores. This protocol holds potential for application to diverse psychiatric conditions.

Publisher's note: Undertaking any experimental protocol requires adherence to local institutional guidelines for laboratory safety and ethics.

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Highlights

Protocol to link neurophysiological features with symptom scores in mental disorders

Workflow for high-density EEG acquisition and preprocessing

Procedures for source localization and functional connectivity network construction

Analytical framework for identifying symptom-associated brain connectomic features

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# Protocol to investigate neurophysiological link between resting-state brain activity and clinical symptoms in methamphetamine use disorder

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## SUMMARY

**The absence of reliable biomarkers impedes the objective diagnosis and monitoring of many psychiatric symptoms. Here, using cue-induced craving in methamphetamine use disorder as a case example, we present a protocol for linking neurophysiological characteristics to clinical symptoms through high-density electroencephalography (HD-EEG). We describe steps for HD-EEG data acquisition, signal preprocessing, source localization, and functional connectivity network construction. We then detail procedures for association analysis with clinical symptom scores. This protocol holds potential for application to diverse psychiatric conditions.**

**For complete details on the use and execution of this protocol, please refer to Tian et al.[1]**

## BEFORE YOU BEGIN

The rising global burden of psychiatric disorders underscores an urgent need for objective biomarkers to improve diagnosis, treatment, and prognosis.[2,3] Current diagnostic methods rely heavily on subjective symptom reports, and the lack of validated neurobiological markers has impeded the development of biomarker-targeted therapies. Emerging evidence, such as the use of resting-state electrophysiological measures for classifying first-episode psychosis, highlights the potential of neurophysiological measures to bridge this gap.[3–5]

High-density electroencephalography (HD-EEG) is a promising tool for this purpose.[6] With its high temporal resolution, enabled by source localization techniques, HD-EEG can reconstruct neural activity with millisecond precision, providing a robust view of the brain's functional networks.[7] Systems with 128 channels or more have been shown to provide higher accuracy in source reconstruction and functional connectivity analysis, making HD-EEG a promising candidate for developing objective biomarkers of psychiatric disorders.[6–8]

This protocol provides a comprehensive, step-by-step framework for utilizing resting-state HD-EEG to identify neurophysiological measures linked to clinical symptoms. It outlines the entire process, from HD-EEG signal acquisition and preprocessing to the computation of functional connectivity networks and symptom correlation. While this protocol is widely applicable across various psychiatric conditions,[1,9,10] we demonstrate the implementation of this approach through a detailed case example of drug cue-induced craving in individuals with methamphetamine use disorder (MUD), a paradigm that has been successfully applied in our previous study.[1,11]

This protocol specifically addresses the challenge of quantifying craving, a core symptom of addiction,[12–14] by employing the Peak Provoked Craving (PPC) strategy.[15,16] This paradigm induces a craving state to facilitate the subsequent investigation of craving-related neurophysiological features. All analytical methods, including power spectrum computation, functional connectivity assessment, and their correlation with clinical symptoms, are designed for maximal generalizability. We established a robust and foundational framework for investigating the neural basis of clinical manifestations, which is applicable to a wide range of psychiatric disorders.[9,10]

## Innovation

This protocol establishes an end-to-end framework that links specific clinical symptoms to underlying neurophysiological features through HD-EEG data analysis in individuals with methamphetamine use disorder and healthy controls. By integrating high-density recording with source-space functional connectivity modeling, the framework enables the construction of fine-grained neural activity maps with improved spatial precision. Compared to previous fragmented pipelines, it provides a unified and reproducible workflow that consolidates all analytical stages across widely used toolboxes (e.g., EEGLAB, FieldTrip, Brainstorm), thereby minimizing inconsistencies arising from cross-platform data transfer. Furthermore, the modular design ensures adaptability to diverse symptom domains and psychiatric populations without compromising analytical rigor. These improvements provide a scalable and generalizable platform for investigating neural mechanisms underlying mental health disorders.

## Institutional permissions

The study was conducted in accordance with the principles of the Declaration of Helsinki and approved by the Institutional Review Board for Human Research at Shanghai Mental Health Center. All data were anonymized, and the privacy of participants was ensured by strict measures.

Please note that research should be initiated only after all necessary institutional approvals have been obtained.

## Preparation

### *Participant recruitment*

**Timing: 1–2 weeks**

Following the experimental design, the standard participant recruitment procedure comprises three main stages.

1. Potential participants identification: Issue a recruitment notice to solicit interested candidates.
2. Eligible participants screening:
   a. Screen for key demographic variables (e.g., age, gender, and educational level).
   b. Administer a structured interview to confirm that each participant meets all inclusion and exclusion criteria.
3. Participants' enrollment:
   a. Outline the study's purpose, procedures, withdrawal protocol, and available support resources.
   b. Execute the informed consent documentation based on the principle of voluntary participation.

*Note:* (1) The inclusion and exclusion criteria must be tailored to the specific research objectives. Generally, the criteria should aim to include key variables that differentiate the experimental groups while excluding confounding factors. (2) In this study, male individuals with methamphetamine use disorder (MUD) were recruited across multiple centers, along with a demographically matched healthy control group. Further participant information can be found in Tian et al.[1]

### *Participant characterization and clinical assessment*

**Timing: 30 min per subject**

4. Characterization collection:
   a. Collect baseline information using a demographic questionnaire.
   b. Assess participant characteristics through standardized questionnaires targeting key domains (e.g., emotional state and sleep quality).

*Note:* Standardized questionnaires include[17–20]: Pittsburgh Sleep Quality Index (PSQI): 19 items (Sleep Quality); Beck Depression Inventory-II (BDI-II): 21 items (Depression Severity); Beck Anxiety Inventory (BAI): 21 items (Anxiety Severity); Barratt Impulsivity Scale-11 (BIS-11): 30 items (Total Impulsivity, with sub-scores for Attention, Motor Impulsivity, and Lack of Planning).

5. Clinical symptom assessment: Evaluate the clinical manifestations of target symptoms.

*Note:* In the current study, participants with MUD were asked to provide the substance use information. Specific data points collected included: onset of drug use, addiction duration, quantity of methamphetamine consumed prior to abstinence, and current abstinence length. The severity of substance addiction was assessed using the scores of the DSM-5 (The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition) questionnaire.[21]

### *High-density EEG acquisition system*

**Timing: About 1.5–2 h**

The system is exceptionally well-suited for investigating the neurophysiological activities related to clinical states.

*Note:* HD-EEG is selected as the primary neuroimaging modality due to its superior temporal resolution, practical applicability in clinical populations, and demonstrated ability to non-invasively map large-scale brain network activity.[6,7]

6. Environment preparation: Select the proper environment for the HD-EEG setup and supplies to seat participants during the recording. Troubleshooting 1.
7. Set up the HD-EEG system:
   a. Hardware connection: Interface the electrode cap with the amplifier, and then establish a link to the recording Terminal.

*Note:* The core hardware, based on Geodesic EEG System 400 (GES 400, Electrical Geodesics, Inc. (EGI)), is listed in the key resources table. A 128-channel saline-based electrode cap (HydroCel GSN, EGI) is chosen for optimal spatial sampling and usability. This cap interfaces with a Net Amps 400 (EGI) amplifier for signal conditioning. Data acquisition and real-time monitoring are performed exclusively within Net Station software (EGI) on a macOS system.

   b. Supply power uniformly to the system with a mobile power source.

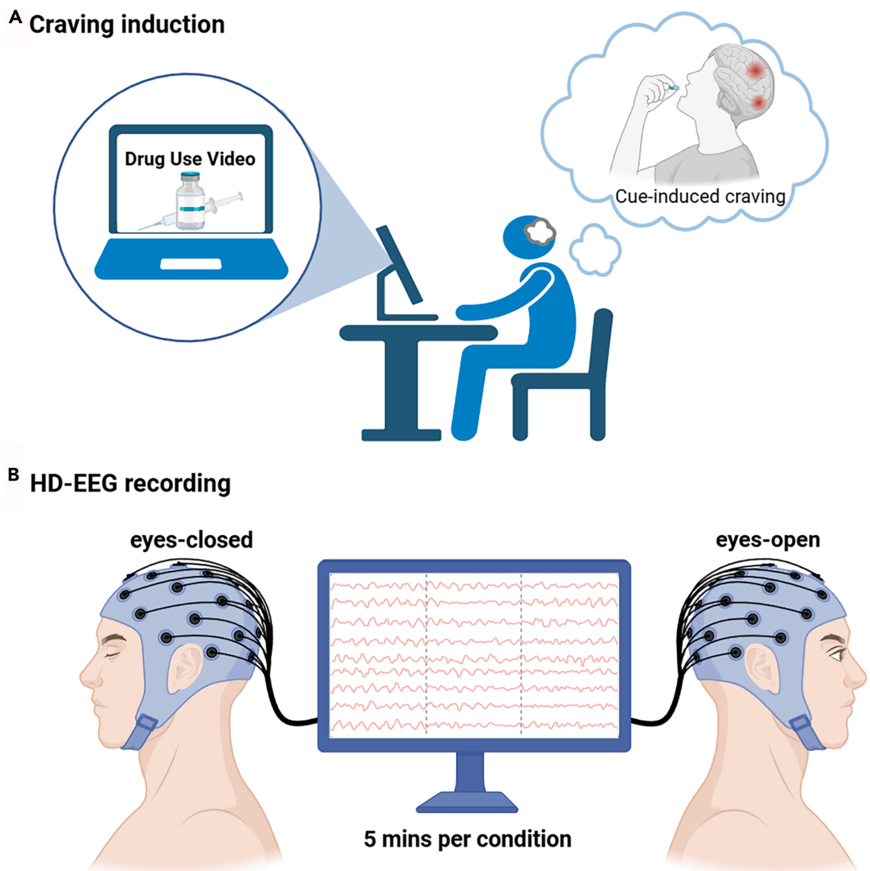



**Figure 1. Experimental paradigm of HD-EEG acquisition**

(A) Craving induction: Craving is induced in individuals with methamphetamine use disorder (MUD) by presentation of a drug-use video.

(B) HD-EEG recording: Five-minute blocks of eyes-open and eyes-closed resting-state HD-EEG data are acquired for each participant.

*Note:* To ensure the integrity of the acquired data, the system utilizes a stabilized power supply to minimize ambient electrical interference.

### *Craving induction stimulus and assessment*

Timing: 10 min per subject

This procedure demonstrates the Peak Provoked Craving (PPC) assessment strategy.

*Optional:* This PPC assessment procedure is mandatory when studying cue-induced craving in MUD cohorts. For studies targeting other psychiatric symptoms, the specific assessment method must be adapted to rigorously quantify the clinical symptoms.

8. Cue-induced craving assessment:
   a. Induce craving: Expose participants with MUD to the drug-use demonstration video (see Figure 1A).

*Note:* A 5-min video stimulus was designed and utilized to establish and quantify robust cue-induced craving states immediately preceding the HD-EEG recording. This video was performed by an individual with MUD, vividly depicting the entire process of methamphetamine use in a naturalistic setting.[12,22]

b. Quantify the subjective craving intensity: Use a Visual Analogue Scale (VAS) to rate participants' current level of craving for methamphetamine on a continuum from 0 (No craving) to 100 (Most intense craving imaginable).

## KEY RESOURCES TABLE

| REAGENT or RESOURCE | SOURCE | IDENTIFIER |
| --- | --- | --- |
| Deposited data | | |
| Data and code | This paper | https://github.com/Weiwen-Tian/EEG_MUD/tree/main |
| Experimental models: Organisms/strains | | |
| MUD Cohort | A total of 162 males over the age of 20, from four drug addiction rehabilitation centers | N/A |
| Health Cohort | A total of 68 males over the age of 20 | N/A |
| Software and algorithms | | |
| Net Station software (version 5.0) | Electrical Geodensics | https://www.magstim.com/product/hd-eeg-ges400/ |
| MATLAB v2021a | MathWorks | https://www.mathworks.com/products/matlab.html |
| EEGLAB version 14.1.1 | Delorme et al.[23] | https://sccn.ucsd.edu/eeglab/index.php |
| FieldTrip version 20180221 | Oostenveld et al.[24] | https://www.fieldtriptoolbox.org/download/ |
| Brainstorm version 3.210507 | Tadel et al.[25] | https://neuroimage.usc.edu/bst/download.php |
| OpenMEEG version 2.4 | Gramfort et al.[26] | https://files.inria.fr/OpenMEEG/download/ |
| Other | | |
| One 128-channel HydroCel GSN EEG recording cap | Electrical Geodensics | https://www.magstim.com/product/hd-eeg-ges400/ |
| One Net Amps 400 amplifier | Electrical Geodensics | https://www.magstim.com/product/hd-eeg-ges400/ |
| One hospital grade isolation transformer | Electrical Geodensics | https://www.magstim.com/product/hd-eeg-ges400/ |
| iMac with Net Station software | Electrical Geodensics | https://www.magstim.com/product/hd-eeg-ges400/ |

## STEP-BY-STEP METHOD DETAILS

### Resting-state HD-EEG recording and preprocessing

**Timing: 30 min per subject**

The following steps are employed to acquire resting-state HD-EEG data and preprocess it through a standardized workflow. It is essential for ensuring the reliability of subsequent analysis results (see Figure 2).

*Note:* After the PPC assessment, high-density resting-state EEG data collection began. This step is crucial to capture the sustained, craving-related brain activity while minimizing interference from task-driven cognitive processes or ongoing sensory perception.

1. Resting-state HD-EEG recording:
   a. System setup and initialization:

*Note:* The power sequence must follow: Mobile power supply → Transformer → Recording computer.

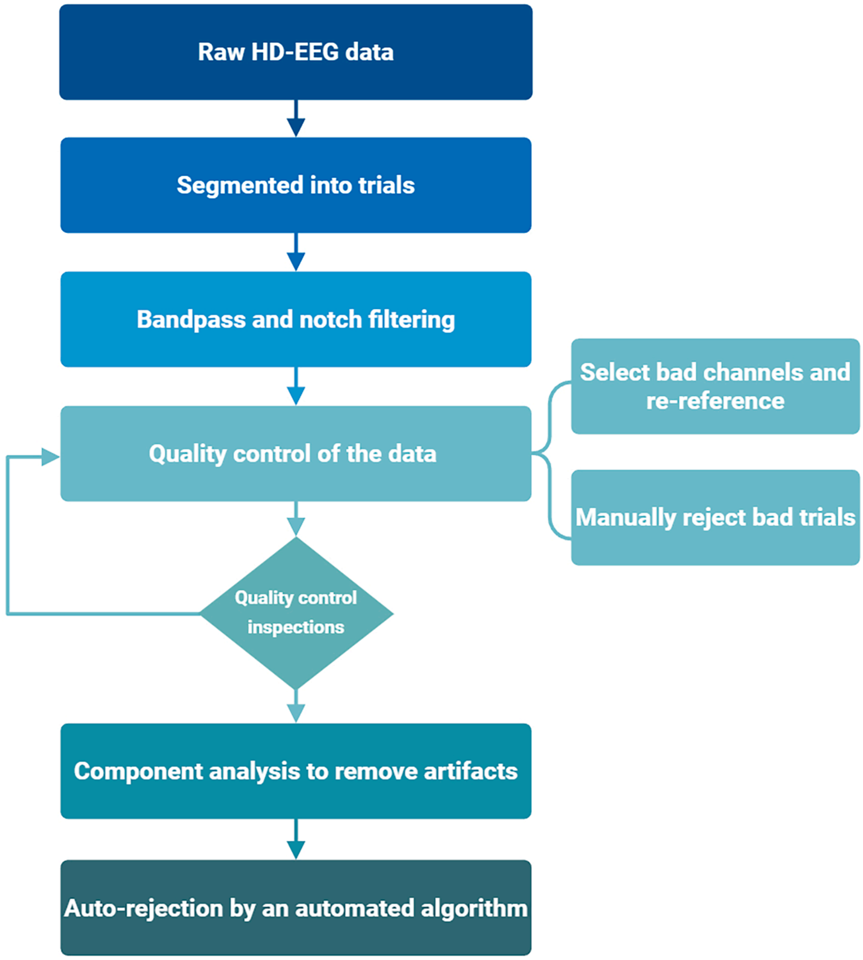



**Figure 2. HD-EEG preprocessing pipeline**

i. The 128-channel saline-based EEG cap is soaked in saline for 20 minutes and then connected to the high-input-impedance amplifier.
ii. Launch the acquisition software and confirm successful signal detection.

b. Participant preparation and impedance check:
   i. Seat the participant in a quiet, low-light environment and instruct them to maintain a relaxed state, refraining from engaging in specific mental tasks.
   ii. Adjust the scalp channels.

⚠ CRITICAL: Maintain electrode impedance values below 50 kΩ by adding saline water to the electrodes.

c. Data acquisition: (see Figure 1B).
   i. Set the sampling rate to 500 Hz and save the final impedance values.
   ii. Record the data in two segments (eyes closed (EC) segment, and eyes open (EO) segment), each 5 minutes in duration. Troubleshooting 2 and 3.

*Note:* Refrain from engaging in specific mental tasks during the recording session.

d. Real-time signal preprocessing and shutdown:
   i. Utilize the built-in notch filter in the acquisition software, set to the local mains frequency (e.g., 50 Hz or 60 Hz), to eliminate line noise.

ii. Save the output files with the '_fil' suffix.
iii. Turn off the system in reverse order of powering on and re-soak the EEG cap for a minimum of 5 minutes before the next subject's preparation.

2. Resting-state HD-EEG data preprocessing:
   a. Data import and segmentation:
      i. Import the raw EEG files (named '_fil') into the EEGLAB environment.
      ii. Segment the continuous resting-state EEG data into epochs of 2 seconds duration.
   b. Sampling rate and filtering:
      i. Downsample the data from the acquisition rate (e.g., 500 Hz) to 250 Hz to reduce computational load.
      ii. Apply an offline band-pass filter: Use a zero-phase Finite Impulse Response (FIR) filter with cutoff frequencies of 1 Hz and 100 Hz.
      iii. Apply a zero-phase notch filter at 50 Hz (or 60 Hz, based on the regional power line frequency) to attenuate residual line noise.

⚠ CRITICAL: Offline Filtering: This deliberate zero-phase filtering (1-100 Hz and notch) is a refinement optimized for advanced analysis and is distinct from the initial filtering performed by the acquisition hardware. The use of zero-phase filters ensures no phase shifts are introduced, which is vital for subsequent functional connectivity analyses.

   c. Electrodes repair:
      i. Manually remove electrode channels exhibiting abnormal overall signals due to poor contact during the recording process.
      ii. Perform spherical interpolation to correct identified bad electrode channels.
   d. Artifact rejection and electrode repair:
      i. Initial trial rejection (visual/threshold): Reject bad trials based on visual inspection of amplitude and waveform characteristics.

*Note:* Use $\pm 3$ standard deviations as a supportive amplitude threshold.

      ii. Automated trial rejection: Apply a trial-detection automated algorithm[27] for auto-rejection based on threshold-based amplitude density and variance.

⚠ CRITICAL: Trials are rejected if the amplitude exceeds 100 μV or the variance surpasses 3 standard deviations from the mean.

   e. Dimensional reduction:
      i. Component identification (ICA): Use Independent Component Analysis (ICA) with the extended information maximization algorithm[28] to identify and model artifacts.

*Note:* Artifacts include eye blinks, eye movements, muscle activity, and other non-brain-related signals.

      ii. Component inspection: Visually inspect the components derived from ICA and manually mark artifactual components.
      iii. Component removal (PCA): Apply Principal Component Analysis (PCA) to remove artifacts through dimensional reduction.

*Note:* The number of dimensions equals the minimum number of principal components that account for more than 99.9% of the total variance.

   f. Re-referencing: Re-reference all electrodes to the average reference.

*Note:* This is the standard procedure for achieving precise source localization and functional connectivity analysis. The initial signal typically employs the Cz electrode positioned at the center of the scalp as the reference point to ensure signal baseline stability in HD-EEG systems utilizing a saline-based matrix.

g. Final data quality check: Visually inspect the fully preprocessed EEG data.

*Note:* It should be confirmed that all artifacts have been successfully removed and that the overall data quality is acceptable for subsequent spectral and functional analysis.

## HD-EEG source localization

**Timing: 1–2 h per subject**

These procedures are designed to convert electrode signals into active source distributions, which are divided into two main components (see Figure 3): solving the forward problem (head model construction) and retrieving the inverse solution (source estimation). Precise source localization facilitates more reliable extraction of neurophysiological features.

*Note:* The entire process utilizes the Brainstorm toolbox (version 3.210507) in MATLAB 2021a.

3. Head model construction:
   a. Anatomical template: Utilize the default ICBM152 template provided within Brainstorm as the anatomical MRI structural image for all participants.

   CRITICAL: While the template is used for standardization, individualized structural MRI data is recommended for achieving higher source localization accuracy.

   b. Electrode registration:
      i. Import the preprocessed EEG data and load the corresponding 128-channel EEG cap model file (obtained from the EGI official website, https://www.egi.com/knowledge-center).
      ii. Edit the MRI registration to ensure the precise positioning of each electrode on the scalp surface. Troubleshooting 4.
   c. Volume head model computation (see Figure 3.bottom): Compute the volume head model using the Boundary Element Method (BEM) via OpenMEEG.

   *Note:* Set the following standard conductivity values for computation: Scalp: 1.0000 S/m; Skull: 0.0125 S/m; Brain: 1.0000 S/m.

4. Source estimation:
   a. Covariance calculation: Compute noise covariance (with no modeling) and data covariance based on the imported preprocessed EEG data in Brainstorm.
   b. Source computation: Estimate the sources using the Minimum Norm Estimation (MNE) approach with the following parameters: Depth Weighting: Order=0.5; Maximal amount=10; Regularization: Regularize noise covariance=0.1. Troubleshooting 5.
5. Source-level data extraction and ROI definition:
   a. Kernel export: Export the kernel data, which maps the channel-level time series to the source level for all subsequent source-level analyses.
   b. Regions of Interest (ROIs) definition: Define the ROIs within the Montreal Neurological Institute (MNI) space.

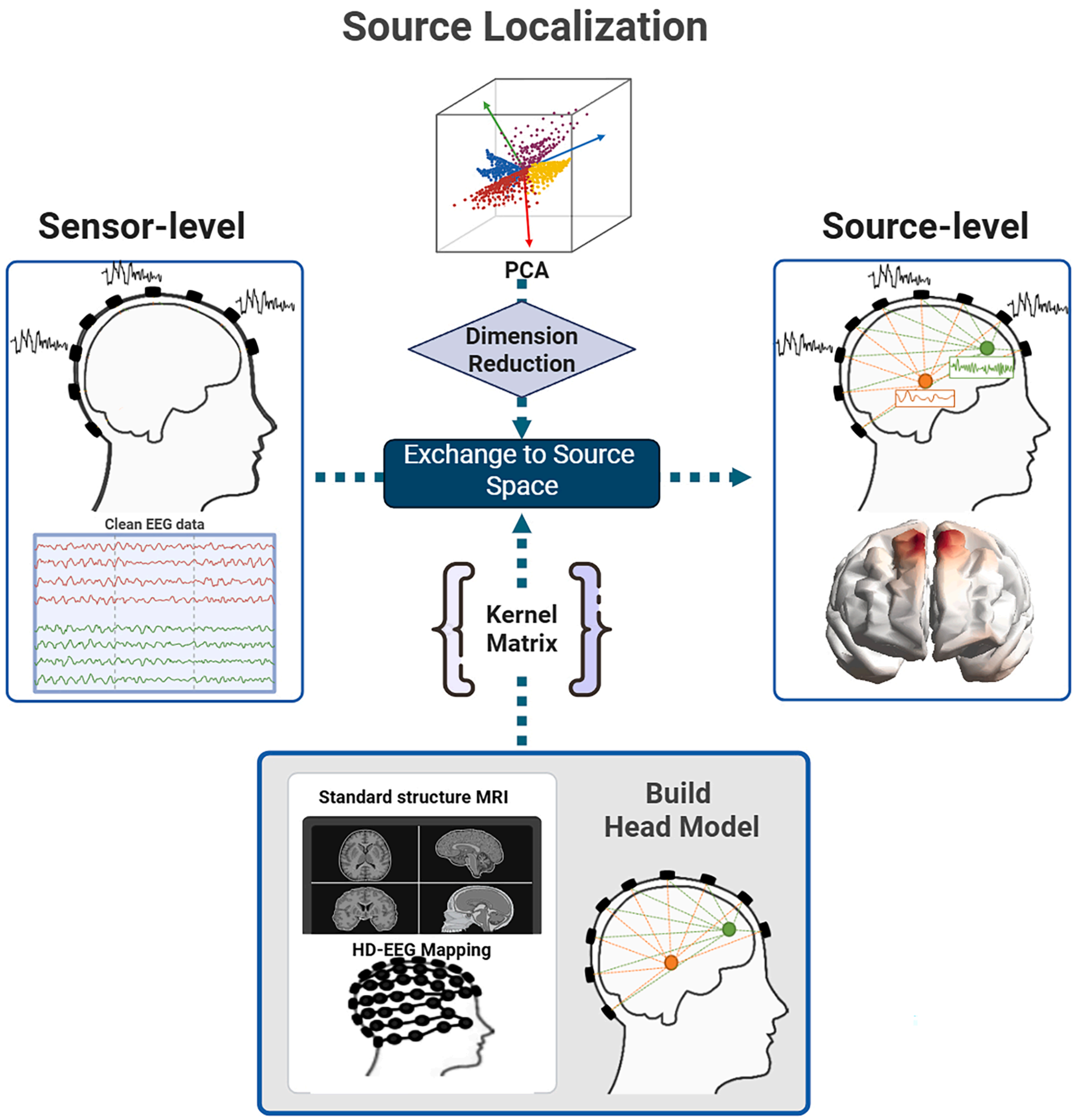



**Figure 3. Flowchart of the source reconstruction pipeline**

The workflow transforms sensor-level HD-EEG signals into source-level activity.

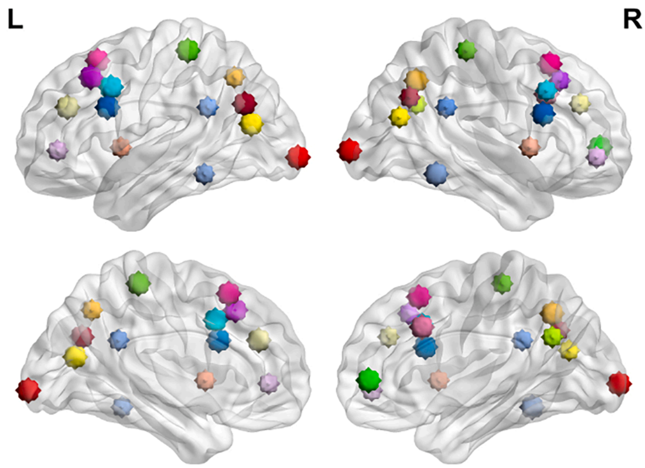
(Bottom) Forward solution: A head model is constructed from a template structural MRI and electrode positions using the boundary element method (BEM). The inverse solution is then used to estimate an imaging kernel at 3,003 cortical dipole sources.

(Middle) Inverse solution and source signal extraction: Sensor-level EEG data are projected to source space via the imaging kernel using the depth-weighted minimum norm estimate (MNE). The resulting source time courses are subsequently reduced in dimensionality using principal component analysis (PCA).

c. Dimensional reduction within ROIs: Apply Principal Component Analysis (PCA) to reduce data dimensions. (see Figure 3.middle).

*Note:* This step captures the main temporal pattern of variation for each ROI by condensing the three-dimensional current density time series at each vertex within an ROI into a single-dimensional representation.

d. Final export: Export the 'crossROI.mat' file, which includes each ROI name and its constituent vertices, for subsequent source-level functional connectivity analysis.




| Visual Network | Somatosensory Network | Dorsal Attention Network | Default Mode Network | Frontoparietal Control Network | Ventral Attention Network |
|---|---|---|---|---|---|
| Visual Area 1 (V1) | Somatosensory Cortex (SMC) | Inferior Frontal Junction (IFJ) | Intraparietal Sulcus (IPS) | Insula (INS) | Dorsal Anterior Cingulate Cortex (DACC) |
| | | Posterior Cingulate Cortex (PCC) | Medial Prefrontal Cortex (MPFC) | Frontal Eye Fields (FEF) | Supplementary Eye Fields (SEF) |
| | | Posterior Middle Frontal Gyrus (PMFG) | Inferior Parietal Lobule (IPL) | Angular Gyrus (ANG) | Middle Temporal Gyrus (MTG) |
| | | Anterior Middle Frontal Gyrus (AMFG) | | Orbital Gyrus (ORB) | Supramarginal Gyrus (SUP) |

**Figure 4. Parcellation scheme of the 31 ROIs**
(Upper) Color-coded spatial distribution of the 31 ROIs on the cortical surface.
(Bottom) Table listing the anatomical labels and their assigned functional network for each ROI. All ROIs are present bilaterally, with the exception of the three underlined regions. Created with BrainNet Viewer.

*Note:* The selection and number of ROIs must be tailored to the specific research question. In this study, focusing on the cortical network dynamics of cue-induced craving, a parcellation of 31 ROIs across 6 major cortical networks was employed.[29,30] (see Figure 4; Table S1) This selection targets networks critically implicated in craving, including the Default Mode, Salience, and Frontoparietal Control networks.[11,31] While this protocol focuses on a cortical framework, the method can be extended to incorporate subcortical ROIs to investigate cortico-subcortical interactions.

## Frequency analysis at the HD-EEG sensor and source level

Timing: About 0.5–1 h per subject

This step describes the method of frequency analysis, deriving the Power Spectral Density (PSD) at both the sensor and source levels (see Figure 5).

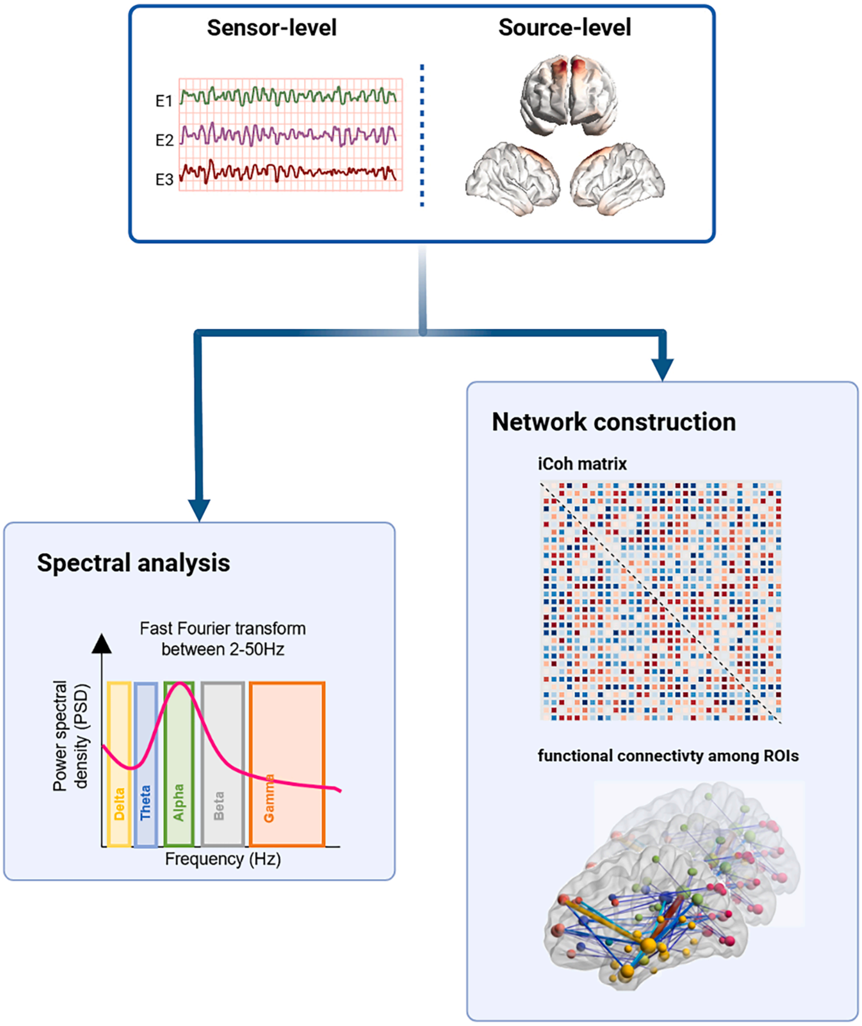



**Figure 5. HD-EEG feature extraction**
(Left) Spectral analysis at sensor and source levels.
(Right) ROI-based functional connectivity network construction using imaginary coherence (iCoh).

*Note:* The subsequent analysis is completed using the FieldTrip toolbox (version 2021) in MATLAB 2021a.

6. Data format conversion and segmentation:
   a. Convert preprocessed data to the standard format within FieldTrip. Troubleshooting 6.
   b. Segment the frequency band into five sub-bands. All subsequent analyses are performed independently within the five standard frequency bands: Delta: 1–3 Hz; Theta: 4–7 Hz; Alpha: 8–12 Hz; Beta: 13–30 Hz; Gamma: 31–50 Hz.
7. Power spectrum estimation (Multi-taper FFT):

*Note:* Estimate the power spectrum of the signals using the multi-taper Fast Fourier Transform (FFT) method with the following parameters: Taper Type: dpss (discrete prolate spheroidal sequences); Taper Smoothing Frequency (tapsmofrq): 1 Hz.

   a. Sensor level: Apply the multi-taper FFT to the preprocessed EEG data.

b. Source level:
   i. Data conversion: Import the imaging kernel matrix from the Brainstorm source localization result and the preprocessed EEG data.
   ii. Compute the transmatrix by converting the channel data to source data via the kernel matrix.
   iii. Apply the multi-taper FFT to the resulting source data (sourcedata) using the transmatrix.

## Functional connectivity network construction of ROIs

**Timing: 1 h per subject**

This section constructs the functional connectivity network by preparing the regional time series data and calculating a robust connectivity metric.

8. Regional time series extraction (ROIdata creation):
   a. Data loading: Load the 'crossROI.mat' file (containing ROI names and vertices) and the kernel data from the source localization stage, along with the sourcedata.
   b. ROI signal mapping: Assign the kernel signals to their corresponding pre-defined Regions of Interest (ROIs), thereby mapping the source activity to specific brain regions.
   c. Representative signal extraction: To derive a representative signal from the multiple kernel time series within each ROI, the Singular-Value Decomposition (SVD) method was adopted.

   *Note:* The first singular vector (a weighted combination of the kernels) is used as the ROI-level signal. This approach optimally weights contributions based on signal strength, providing a functionally unified, robust estimate (superior to a simple arithmetic mean)[32] and enhancing interpretability.

9. Frequency domain processing of ROI data:
   a. Apply the multi-taper Fast Fourier Transform (FFT) method to process the extracted ROI data with updated parameters suitable for connectivity analysis: Taper Type: dpss; Taper Smoothing Frequency (tapsmofrq): 1.9 Hz. (see Figure 6).

   *Note:* The 1.9 Hz half-bandwidth parameter is the default setting in FieldTrip, offering an optimal balance between spectral smoothing, noise reduction, and frequency resolution for lower-frequency phase relation analysis. For high-frequency signals (e.g., high-gamma), this half-bandwidth should be increased accordingly to capture broader spectral features.

10. Connectivity metric calculation and FCN construction:
    a. Perform pairwise computation of imaginary coherence between all selected ROIs to construct a functional connectivity matrix.

    *Note:* Imaginary coherence: Imaginary coherence is a preferred metric for source-level EEG analysis due to its inherent robustness to volume conduction. This provides superior specificity for genuine neural interactions, facilitating the discovery of objective biomarkers.[33] While Phase-Lag Index (PLI)[34] and Weighted PLI (wPLI)[35] are valid alternatives, conventional measures like Phase-Locking Value (PLV) and Mutual Information (MI) are discouraged as they are sensitive to zero-lag synchronization.[36]

11. Average the functional connectivity matrices of all individuals in the group to obtain the functional connectome results for experimental groups (in this study, MUD and HC, respectively).

```
%% Compute functional connectivity
% Configuration settings for FC analysis
        cfgcoh              = [];
        % choose the method of time series analysis
        cfgcoh.method       = 'mtmfft';
        cfgcoh.pad          = 'nextpow2';
        % set the taper-type
        cfgcoh.taper        = 'dpss';
        % set the output type
        cfgcoh.output       = 'fourier';
        % cfg.foi contains five frequency bands
        cfgcoh.foi          = cfg.foi;
        cfgcoh.channel      = 'all';
        cfgcoh.keeptrials   = 'yes';
        cfgcoh.tapsmofrq    = 1.9;
        % functional connectivity indicator
        cfgcc.method        = 'coh';
        cfgcc.complex       = 'imag';
% Run FC analysis
        powcsd = ft_freqanalysis(cfgcoh, Data);
        tempcoh = ft_connectivityanalysis(cfgcc, powcsd);
```

**Figure 6. Parameter configuration code for functional connectivity computation**
Exported from Matlab 2021a.

## Connectome statistical analysis

Timing: About 1–2 h

The following protocol gives guidance on the constitution of a statistical pipeline, involving a generalizable workflow for linking neurophysiological data to clinical outcomes. The primary output is a set of statistically significant functional connections, along with their corresponding effect sizes and statistical values. Troubleshooting 7.

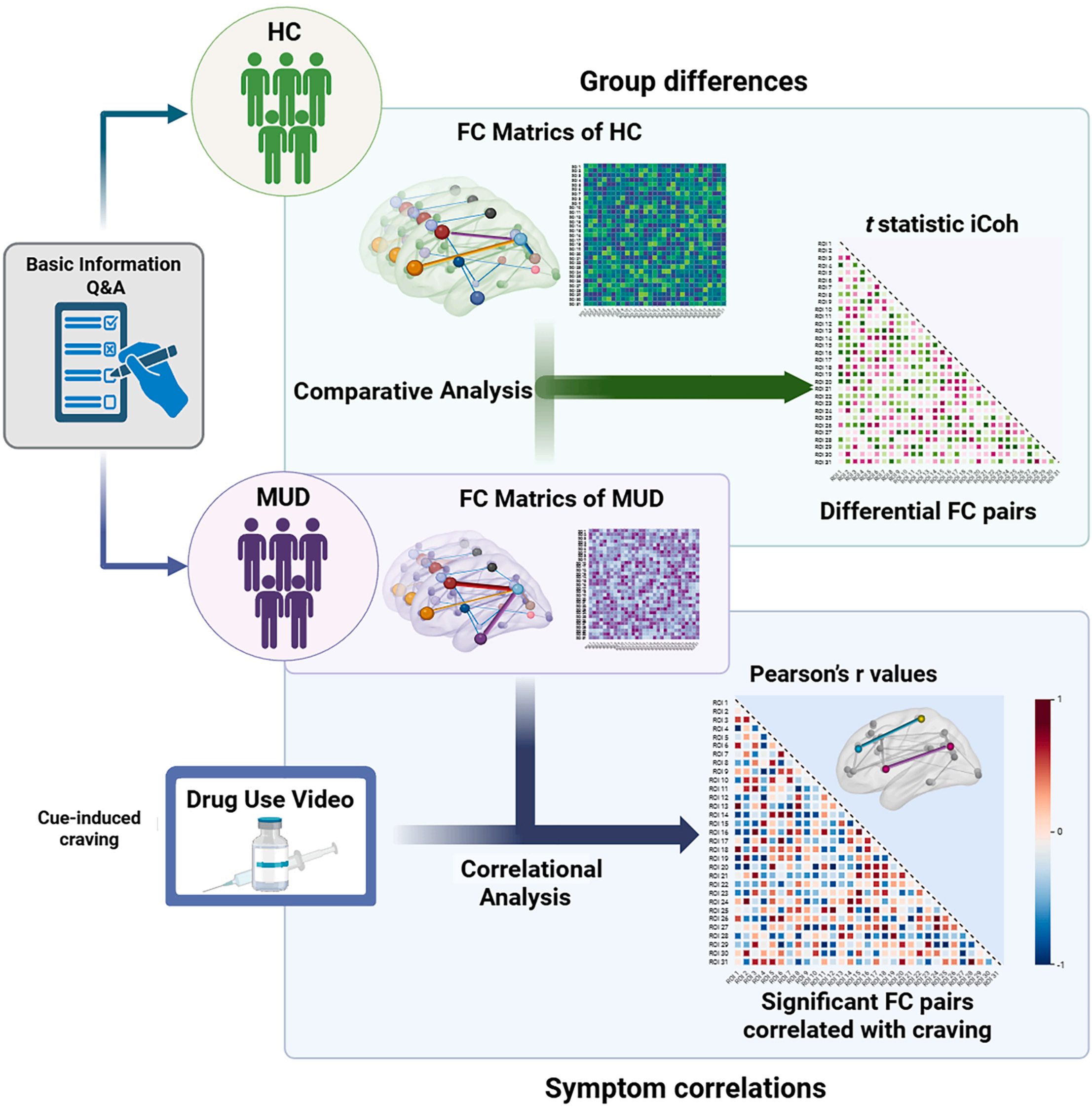



**Figure 7. Statistical analysis workflow for functional connectomes**
(Upper) Inter-group comparison: Identification of connections with significant group differences using t-tests, FDR-corrected.
(Bottom) Symptom correlation: Identification of connections significantly correlated with a clinical symptom within the target cohort, after multiple comparisons correction.

12. Identification of group-differentiating functional connectome: (between-group differences, see Figure 7.upper).
    a. Group comparison:
        i. Perform independent samples t-tests on each functional connection to compare the clinical and control groups.
        ii. Record the t-statistic and raw p-value for every connection.
    b. Correction: Apply the Benjamini-Hochberg false discovery rate (FDR)[37] procedure for multiple comparisons to correct all raw p-values, yielding corresponding q-values.
    c. Output: Define connections that survive the FDR correction ($q < 0.05$) as the significant group-differentiating connectome alterations.
13. Identification of symptom-correlated connectivity features: (see Figure 7.bottom).

a. Confound control: Screen ancillary behavioral variables, incorporating those significantly correlated with the primary clinical symptom score (e.g., craving) as covariates.
b. Correlation: Calculate the Pearson correlation coefficient between the primary symptom score and the strength of each functional connection across all participants in the target group.

*Note:* This yields a correlation coefficient (r) and an associated *p* value for each connection.

c. Correction: Submit all p-values to the Benjamini-Hochberg FDR procedure for multiple comparisons correction.
d. Output: Classify connections with an FDR-corrected q-value < 0.05 as significant symptom-related connectivity features.

## EXPECTED OUTCOMES

This protocol establishes a systematic methodological framework for investigating the neurophysiological signatures of craving within clinical populations. The workflow integrates three core phases: HD-EEG source localization, functional connectivity network construction, and the feature connectome extraction.

Designed for group-level statistical analysis, following the proposed pipeline process will ultimately generate two primary outcomes: (1) Symptom-specific functional connectomes that statistically differentiate clinical cohorts (MUD vs. controls); and (2) Connectivity features that significantly correlate with specific clinical and behavioral measures (e.g., craving severity). These outcomes reveal the abnormal neurophysiological characteristics of the disease and their rigorous, systematic association mapping with specific clinical symptoms.

By providing an objective, source-level analytic framework, this protocol addresses the persistent challenge of objective biomarkers in diagnosis and treatment. The robustness of the core steps demonstrates broad potential for extension and validation across a wide range of other psychiatric and neurological disorders.

## LIMITATIONS

While our protocol provides a solid framework, its application has some constraints. The workflow is exclusively optimized for resting-state EEG and lacks methods for necessary task-state analysis.[38] Another major limitation is the sample. This severely restricts generalizability and risks bias from sex differences and institutional effects.

It will be necessary to validate this protocol in diverse cohorts (e.g., women, outpatients) through multi-site collaboration. Finally, our reliance on correlation to discover brain-symptom links is deficient in explaining causation. The next step is to integrate machine learning models[10,39,40] to predict symptoms, thereby strengthening the evidence for a functional link. Despite these limits, the protocol retains significant potential for mapping neurophysiological features and functional connectomes across other psychiatric disorders.

## TROUBLESHOOTING

### Problem 1

Electromagnetic interference (related to preparation Step 6).

The HD-EEG system is susceptible to environmental electromagnetic interference, which in turn generates excessive data noise.

### Potential solution

- Conduct experiments of the HD-EEG system in a dedicated and shielded environment.
- Ensure participants remove all personal electronic devices and metallic items prior to recording.

### Problem 2

Motion artifacts (related to Step 1.c.ii).

The dense recording electrode array is highly sensitive to the participants' body movements. Subjects should be instructed to remain as still as possible throughout the recording process.

### Potential solution

- Prior to the recording, explicitly inform participants that excessive movement would create artifacts, which potentially necessitate a re-recording.
- Monitor the participant's state continuously during the recording and record when movement occurs in the process.

### Problem 3

Unstable impedance (related to Step 1.c.ii).

The 128-channel saline-based electrode cap demonstrates relatively short impedance stabilization durations, which makes it a challenge to maintain electrode impedance values below 50 kΩ.

### Potential solution

- Instruct participants to wash their hair before testing.
- Fully immerse all 128 electrodes in saline solution for 20 minutes.
- Ensure optimal electrode-scalp contact during cap placement.
- Recheck and adjust impedance values after each recording session.

### Problem 4

Inaccurate co-registration (related to Step 3.b.ii).

### Potential solution

- Use individualized structural MRI for co-registration where available. If using a template, ensure compatibility with the electrode cap.
- Precisely align anatomical landmarks (e.g., nasion, pre-auricular points).
- Visually inspect co-registration residuals and source-electrode alignment.

### Problem 5

Suboptimal reconstruction parameters (related to Step 4.b).

### Potential solution

- Optimize regularization and depth weighting parameters on a data subset.
- Apply cross-validation to balance spatial specificity and noise suppression.
- Compare outcomes across noise covariance estimates to ensure robustness.

### Problem 6

Invalid data structure (related to Step 6.a).

The input data that does not adhere to the required structural specifications cannot be properly analyzed in FieldTrip.

### Potential solution

- Data conversion: Use the various built-in conversion programs in the toolbox to perform data conversion. For basic format conversions, utilize ft_datatype_raw or ft_datatype_timelock; for external data formats, employ dedicated adapters (e.g., fieldtrip2eeglab for EEGLAB).
- Data check: Apply ft_checkdata for structural validation.

### Problem 7

Trade-off between statistical power and false positives (related to steps 12 and 13).

Large-scale connectome analyses face a trade-off: stringent methods reduce power, while lenient approaches increase false positives.

### Potential solution

- Perform a priori power analysis to determine sample size.
- Limit multiple comparisons by selecting ROIs based on hypotheses and reducing dimensionality (e.g., using the first principal component).
- Choose appropriate correction methods: FDR for ROI-based analyses; permutation testing for whole-brain connectivity.

## RESOURCE AVAILABILITY

### Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Ti-Fei Yuan (ytf0707@126.com).

### Technical contact

Specific technical questions on executing this protocol should be directed to and will be answered by the technical contact, Yi-Han Yan (y9946h@sjtu.edu.cn).

### Materials availability

This protocol did not generate new unique reagents. The video material for inducing craving is available upon reasonable request from the lead contact.

### Data and code availability

This protocol was mainly based on the published study by Tian et al.,[1] which reported all related data and used codes. Any additional information is available from the lead contact upon request.

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## AUTHOR CONTRIBUTIONS

Y.-H.Y. drafted and revised the manuscript. Y.Z., T.H., and X.L. critically revised the manuscript. D.Z. supervised the overall manuscript revision. T.-F.Y. and D.Z. designed the study protocol and secured funding.

## DECLARATION OF INTERESTS

The relevant protocol procedure is protected by an issued Chinese invention patent (no. CN117807370A) and a published PCT application (no. WO/2025/102844).

## SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.xpro.2025.104276.

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